MIL-C-43390C

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SUPERSEDING
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MILITARY SPECIFICATION

CLIPPER, HAIR, AND BLADE SETS (SMALL ANIMAL)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

* 1.1 Scope. This specification covers an electrically operated small animal hair clipper and three sizes of interchangeable blade sets.

* 1.2 Classification. The hair clipper and the blade sets shall be the following items as specified (see 6.2).

Item 1 - Hair clipper with size 10 blade set and accessory kit
Item 2 - Blade set, size 10 - General clipping
Item 3 - Blade set, size 15 - Medium clipping
Item 4 - Blade set, size 40 - Surgical clipping

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

PPP-B-566 - Boxes, Folding, Paperboard.
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-B-676 - Boxes, Set-Up, Paperboard.

MILITARY

MIL-P-116 - Preservation, Methods Of.

FSC 3770

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply:

National Motor Freight Traffic Association, Inc., Agent

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street, N.W., Washington, DC 20036.)

Uniform Classification Committee, Agent

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

Underwriters' Laboratories, Inc. (UL)

UL 73 - Motor-Operated Appliances

(Application for copies of UL publications should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1285 Walt Whitman Road, Melville, Long Island, NY 11746; or 1655 Scott Boulevard, Santa Clara, CA 95050.)

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American Society for Testing and Materials (ASTM)

E 18 - Methods of Test for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Standard product. The hair clipper and blade sets delivered under this specification shall be the manufacturer's commercial product, except for changes necessary to comply with specification requirements. All like items furnished on any one contract, including parts and subassemblies thereof shall be new and interchangeable.

* 3.2 Codes and standards. The hair clipper shall comply with the applicable requirements of UL Standard No. 73.

* 3.2.1 Compliance. Prior to the approval of the first article, if one is submitted, or prior to commencing production, the supplier shall submit to the contracting officer or his authorized representative satisfactory evidence that the hair clippers conform to the applicable requirements of UL Standard No. 73.

* 3.2.1.1 UL. Acceptable evidence of meeting the requirements of UL shall be the UL label, a UL listing mark, or a certified test report from a recognized independent testing laboratory, acceptable to the Government, indicating that the hair clippers have been tested and conform to UL Standard No. 73.

* 3.3 Materials and components. Materials and components shall be as specified herein. Materials not definitely specified shall be of the quality normally used by the manufacturer in his standard commercial hair clippers and blade sets provided the completed items comply with provisions of this specification.

* 3.4 Design and construction.

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* 3.4.1 Item 1, hair clipper with size 10 blade set. The hair clipper shall be electrically operated and consist of a body, electric motor, wiring circuit, on-off switch, security line fitting, detachable cutter head, power cord, blade set guard, a size 10 blade set, the necessary assembly hardware and the accessory kit specified in 3.4.1.6. When fully assembled the clipper shall weigh not more than 1-1/2 pounds. The length of the fully assembled clipper, exclusive of the power cord, shall be not more than 9 inches. When tested as specified in 4.4.1, the clipper shall operate without evidence of mechanical failure or of overheating. When drop tested as specified in 4.4.2, the security line fitting shall not break, pull loose from the clipper body, or become permanently deformed and the clipper shall operate without evidence of damage.

* 3.4.1.2 Body. The body shall be fabricated of thermosetting, phenolic plastic in accordance with the manufacturer's standard commercial practice and shall contain the electric motor, the wiring circuit and the on-off switch. The power cord and the security line fitting shall be attached to the body. Means of bearing lubrication through the body shall be provided.

* 3.4.1.3 Electrical characteristics. Unless otherwise specified (see 6.2), the clipper shall operate on nominal 115 volts, and the motor shall operate on 50-60 hertz or direct current power supply and shall draw not more than 45 watts. The power cord shall be not less than 10 feet in length and shall conform to UL requirements for type SVO, ST, SJT, or SJO.

* 3.4.1.4 Blade set guard. The blade set guard shall be of rubber or of plastic and shall fit over the blade set when mounted in the cutter head and shall protect the teeth and blade edges. The guard shall be held in place by friction and shall be removable and replaceable without the use of tools.

* 3.4.1.5 Cutter head. The cutter head housing shall be fabricated of thermosetting phenolic plastic in accordance with the manufacturer's standard commercial practice and shall contain the blade set drive mechanism and shall be removable and replaceable on the body without the use of tools. The blade set drive mechanism shall be covered by a removable corrosion-resistant metal plate held in place by not less than two screws.

* 3.4.1.6 Accessory kit. Each clipper shall be furnished with an accessory kit consisting of the following items.

1 - Tube of lubricating oil, 1/2 ounce
1 - Tube of grease
1 - Set of electric motor brushes
1 - Set of electric motor brush springs
1 - Set of blade clamping springs with mounting screws
1 - Blade set mounting screw

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* 3.4.2 Blade sets, items 2, 3, and 4. Each blade set shall consist of a top and a bottom blade of polished tool or carbon steel, a blade clamping spring, and two clamping spring mounting screws. The blade set shall be held in place in the cutter head by a blade set mounting screw. The head of the blade set mounting screw and the ends of the clamping spring mounting screws shall not protrude beyond the bottom face of the bottom blade. The bottom blade shall be stationary and cutting shall be accomplished by the lateral movement of the top blade over the bottom blade. The blade teeth shall have a hardness of from 58 - 68 on the Rockwell C scale when tested as specified in 4.4.3. The different size blade sets shall be interchangeable in the cutter heads with a screw driver.

* 3.4.2.1 Item 2, size 10, general clipping blade set. The bottom blade of the size 10 blade set shall have 26 full teeth and 2 end half teeth. The top blade shall have 18 full teeth. The blade thickness of the bottom blade at the point of first cut shall be not greater than 0.0787 inch.

* 3.4.2.2 Item 3, size 15, medium clipping blade set. The bottom blade of the size 15 blade set shall have 26 full teeth and 2 end half teeth. The top blade shall have 18 full teeth. The blade thickness of the bottom blade at the point of first cut shall be not greater than 0.0394 inch.

* 3.4.2.3 Item 4, size 40, surgical clipping blade set. The bottom blade of the size 40 blade set shall have 41 full teeth and 2 end half teeth. The top blade shall have 25 full teeth. The blade thickness of the bottom blade at the point of first cut shall be not greater than 0.0197 inch.

* 3.5 Finish. The hair clipper and blade sets shall be finished in accordance with the manufacturer's standard commercial practice.

* 3.6 Marking for identification. Each hair clipper and blade set shall be permanently and legibly marked with the manufacturer's name, trade name or trademark of such known character readily identifiable with the manufacturer. In addition, each blade set shall be marked with the blade set size and the model number of the clipper with which they are intended to be used.

* 3.7 Workmanship. All components and assemblies of the hair clippers and blade sets shall be free from dirt and other harmful extraneous material, burrs, slivers, rough die, tool and grind marks, dents and cracks. Castings, molded parts and stampings, if used, shall be free of sand, fins, pits, blow holes and sprues. External surfaces, except for teeth on blade sets, shall be free from sharp edges and corners. Blade sets shall have no broken, missing or malformed teeth and teeth shall be uniformly spaced.

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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

4.2 Quality conformance inspection. Except as otherwise specified herein, sampling for inspection shall be performed in accordance with the provisions set forth in MIL-STD-105.

4.2.1 Component and material inspection. In accordance with 4.1 above, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

* 4.2.2 End item inspection. The inspection lot shall be all hair clippers or blade sets of one size offered for inspection at one time. The sample unit for this inspection shall be one hair clipper, or one blade set, as applicable.

* 4.2.2.1 Visual examination. The hair clippers and the blade sets, as applicable, shall be examined for the defects in table I. The inspection level shall be II with an AQL (acceptable quality level) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

* TABLE I. Classification of defects

Examine	Defect	Classification Major	Minor
Item 1			
Standard product	Not manufacturer's commercial product except for changes necessary to comply with specification requirements	X	

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* TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Item 1 (cont'd)			
Components	Any component missing or not as specified	X	
Finish	Not finished in accordance with manufacturer's standard commercial practice		X
	Dirty extraneous material not removed		X
Cutter head	Not removable and replaceable without use of tools	X	
On-off switch	Missing, not mounted on body	X	
Blade set guard	Missing, does not cover blade teeth and edges. Requires use of tools to remove or replace	X	
Accessory kit	Any item missing	X	
Lubrication	Lubrication means not provided as specified	X	
Workmanship	Burr, sliver, rough die, tool mark, grind mark		X
	Dent, crack, sharp edge or corner	X	
Castings, mouldings and stampings	Sand, fins, pits, blow holes, sprues	X	
Marking	Missing, not as specified	X	
Items 2, 3 and 4			
General	Not interchangeable	X	
	Requires tools other than screw driver to remove or install	X	
	Any component missing or not as specified	X	

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* TABLE I. Classification of defects (cont'd)

Examine	Defect	Classification Major	Minor
Items 2, 3 and 4 (cont'd)			
Finish	Top and bottom blades not polished steel	X	
Size	Not size specified	X	
	Size not marked on each blade set	X	
Teeth	Not number specified	X	
	Broken, malformed, missing, not uniformly spaced	X	

* 4.2.2.2 Dimensional and weight examination. The hair clipper and the blade sets, as applicable, shall be examined for compliance with dimensions and weight specified. Any dimension or weight not within the specification tolerance shall be a defect. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

* 4.2.2.3 End item testing. Each sample hair clipper or blade set, as applicable, shall be tested as specified in 4.4.1, 4.4.2, and 4.4.3, as applicable. Failure of a test shall be cause for rejection of the inspection lot. The inspection level shall be S-2.

4.3.3 Inspection of preparation for delivery. An examination shall be made to determine that preservation, packaging, packing, and markings are in compliance with section 5. Defects shall be as specified in table II. The sample unit for this examination shall be one shipping container fully prepared for delivery. The lot shall be the number of containers submitted for inspection at one time. The inspection level shall be S-2 with an AQL of 2.5, expressed in terms of defects per hundred units.

TABLE II. Examination of preparation for delivery

Examine	Defect
Marking	Omitted; incorrect; illegible; of improper size, location, sequence or method of application
Preservative	Improperly applied or missing
Weight	Net weight is more than specified

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4.3.4 Examination of shipping containers. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in closure, waterproofing, and strapping shall be in accordance with the appendix of the applicable specification.

* 4.4 Test methods.

* 4.4.1 Operational test. Each sample clipper, fully assembled, shall be connected to a compatible electric power source and allowed to operate for not less than 5 minutes. The clipper shall be examined during the 5 minute operating period to determine compliance with the mechanical failure and overheating requirements of 3.4.1. Noncompliance with specified requirements shall constitute failure of this test.

* 4.4.2 Drop test. A security line not less than 3 feet long shall be attached to the security line fitting and to a fixed anchor. The clipper shall be turned on. The clipper shall then be allowed to drop in a free unobstructed fall for a distance of not less than 3 feet to the end of the security line. The clipper shall then be examined to determine compliance with the drop test requirements of 3.4.1. Noncompliance with specified requirements shall constitute failure of this test.

* 4.4.3 Hardness test. Tests for hardness of cutting blades shall be performed in accordance with ASTM E 18 to determine compliance with the hardness requirements of 3.4.2. Cutting edge hardness shall be taken on each end tooth and the middle tooth. Hardness reading shall be taken in the center of the widest surface area of the teeth. Nonconformance with specified requirements shall constitute failure of this test.

5. PREPARATION FOR DELIVERY

* 5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

* 5.1.1.1 Preservation. The blade sets shall be cleaned, dried, and individually prepared in accordance with method I of MIL-P-116, using a commercial lubricant containing a rust inhibitor.

* 5.1.1.2 Prepackaging. Each blade set, cleaned and preserved as specified in 5.1.1.1, shall be packaged in a snug-fitting folding paperboard box conforming to variety 1, style II, type D, class a of PPP-B-566. The accessory kit shall be packaged in accordance with the manufacturer's standard practice.

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* 5.1.1.3 Packaging. The cord attached to the clipper shall be compactly coiled and tied by suitable means. Each complete clipper (item 1) shall then be packaged in a snug-fitting folding or setup paperboard box conforming to variety 1, style III or VIII, type G, class 1 of PPP-B-566 or type I, variety 1, class A, style 3 or 4 of PPP-B-676. The contents of each box shall be cushioned by suitable means to prevent movement or damage while in transit. Twelve blade sets of one size only, preserved and prepackaged as specified in 5.1.1.1 and 5.1.1.2, respectively, shall be packaged in paperboard boxes as specified herein.

* 5.1.2 Level C. The hair clippers and blade sets shall be preserved and packaged to afford adequate protection against corrosion, deterioration, and damage during shipment from the supply source to the first receiving activity. The supplier may use his standard practice when it meets these requirements.

* 5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

* 5.2.1 Level A. Twelve hair clippers, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, V3s of PPP-B-636. Each shipping container shall be closed, waterproofed by means of tape, and reinforced with flat strapping or tape banding in accordance with PPP-B-636. Blade sets of one size only, preserved and packaged as specified in 5.1, shall be packed as specified herein except the weight of contents of each shipping container shall not exceed 40 pounds.

* 5.2.2 Level B. Twelve hair clippers, preserved and packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard shipping container conforming to style RSC, type CF, variety optional or type SF, class domestic, grade 275 of PPP-B-636. Each shipping container shall be closed in accordance with method II of the appendix of PPP-B-636. Blade sets of one size only, preserved and packaged as specified in 5.1, shall be packed as specified herein except the weight of contents shall not exceed 65 pounds. When specified (see 6.2), the shipping container shall be V3c, V3s, or V4s, fabricated and closed in accordance with PPP-B-636.

* 5.2.3 Level C. Hair clippers and blade sets preserved and packaged as specified in 5.1, shall be packed in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.3 Marking. In addition to any special marking required by the contract or order, intermediate packages and shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

* 6.1 Intended use. The hair clippers, item 1, covered by this specification are intended for use in cutting hair on small animals. Blade set sizes and their intended use are as follows:

Item 2 - Size 10 - General and under body clipping
Item 3 - Size 15 - Medium-close clipping (mane, fetlock, udder, and cat)
Item 4 - Size 40 - Surgical area clipping

* 6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Item or items required (see 1.2).
(c) When required electrical characteristics are other than as specified (see 3.4.1.3).
(d) Selection of applicable levels of preservation and packaging, and packing (see 5.1 and 5.2).
(e) Whether weather-resistant class fiberboard shipping containers are required for level B shipment (see 5.2.2).

* 6.2.1 Contract data requirements. Any data items to be delivered under any contract for items covered by this specification should be specifically called for in the contract in accordance with the applicable regulations of the procuring activity.

6.3 The margins of this specification have been marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

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